Dirac Particles in the Spherical Basis

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Vacuum
Gauge
Electrodynamics

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1 Review of Spherically Based Four-Coordinates

The instantaneous position $\vec{r_s}$ of an elementary particle moving along its world line is indicated in the spacetime diagram of Figure 1. An arbitrary vector $\vec{r}$ is causally connected to the motion of the particle by writing

$$\vec{R} \equiv \vec{r} - \vec{r_s} \tag{1.1}$$

The vector $\vec{R}$ can represent the worldline of any particle emitted at the retarded time ct_r—subject to the requirement that it moves at the speed of light. Photons and Weyl neutrinos are good candidates, but any high energy emission of mass $m \neq 0$ will work to a good approximation. Now suppose the vectors $\vec{R}$ and $\vec{r_s}$, are decomposed in terms of their timelike and spacelike components,

$$\vec{R} = (\vec{R} \cdot \vec{\beta})\vec{\beta} - (\vec{R} \cdot \vec{U})\vec{U} = \rho[\vec{\beta} + \vec{U}] \tag{1.2a}$$

$$\vec{r_s} = (\vec{r_s} \cdot \vec{\beta})\vec{\beta} - (\vec{r_s} \cdot \vec{U})\vec{U} = (c\tau - \rho)\vec{\beta} \tag{1.2b}$$

where $c\tau$ is elapsed time in the rest frame of the particle. Inserting both equations (1.2) into (1.1) and solving for $\vec{r}$ determines

$$\boxed{\vec{r} = c\tau\vec{\beta} + \rho\vec{U}} \tag{1.3}$$

Although not readily apparent, this vector relation qualifies as a coordinate transformation from a Cartesian set of coordinates to a spherically based system with components of $\vec{r}$ satisfying

$$r^\nu = (ct, x, y, z) = f(c\tau, \rho, \theta, \phi) \tag{1.4}$$

The two angles (θ, ϕ) are defined by writing the spatial components of $\vec{R}$ in terms of spherical polar coordinates

$$\boldsymbol{R} = R\hat{\boldsymbol{n}} = (R\sin\theta\cos\phi, R\sin\theta\sin\phi, R\cos\theta) \tag{1.5}$$

Basis Vectors: Basis vectors $\vec{\mathcal{S}}_\alpha$ associated with equation (1.3) are listed in the first column of Table 1. This column also indicates the set of unit vectors

$$\vec{\beta} = \beta^\nu \vec{\mathrm{e}}_\nu \qquad \vec{U} = U^\nu \vec{\mathrm{e}}_\nu \qquad \vec{\vartheta} = \theta^\nu \vec{\mathrm{e}}_\nu \qquad \vec{\varphi} = \phi^\nu \vec{\mathrm{e}}_\nu \tag{1.6}$$

with explicit forms of each one displayed in the third column of the table. There are six orthogonality relations among the pairs of four-vectors. Additionally, the timelike vector has a magnitude of $+1$, while the spacelike vectors have magnitudes of -1 for easy construction of the flat space metric tensor. It's also important to indicate formulas for angular four-vectors by differentiating $\vec{U}$:

$$\frac{\partial \vec{U}}{\partial \theta} = \frac{R}{\rho} \cdot \vec{\vartheta} \qquad \frac{\partial \vec{U}}{\partial \phi} = \frac{R}{\rho}\sin\theta \cdot \vec{\varphi} \tag{1.7}$$

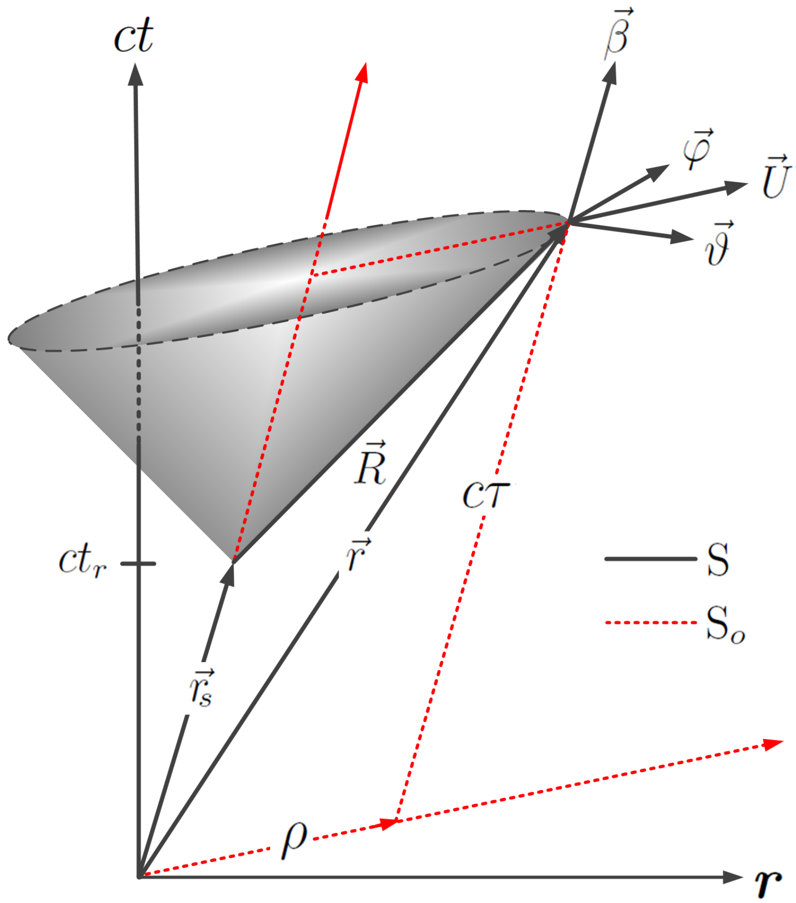

Figure 1: *Orthonormal set of unit four-vectors at the point $\vec{r}$ defined relative to the retarded position of a particle moving along its world line. All of equations (1.1 - 1.3) can be determined by inspection of the diagram.*

Inverse: An important property of the transformation in equation (1.3) is the existence of an inverse. While there are similarities to the inverse transformation in 3-space spherical-polar coordinates, the four-space transformation is somewhat more complicated, and may be written

$$c\tau = ct_r/\gamma + \rho \qquad \rho = \gamma R - \gamma \boldsymbol{R}\cdot\boldsymbol{\beta} \tag{1.8a}$$

$$\theta = \tan^{-1}\left[\frac{(R_x^2 + R_y^2)^{1/2}}{R_z}\right] \qquad \phi = \tan^{-1}\left[\frac{R_y}{R_x}\right] \tag{1.8b}$$

Although the cartesian coordinates are clearly not visible here, they can be found buried within the components (ct_r, R_x, R_y, R_z). Basis vectors from the inverse transformation are labeled $\vec{\jmath}^{\nu}$ and are displayed in the center column of Table 1.

$\vec{\mathcal{S}}_\nu$	$\vec{\mathcal{J}}^\nu$	Unit Vector
$\vec{\mathcal{S}}_\tau = \dfrac{\partial \vec{x}}{\partial c\tau} = \beta^\nu \vec{\mathrm{e}}_\nu$	$\vec{\mathcal{J}}^\tau = \partial_\nu c\tau \vec{\omega}^\nu = \beta_\nu \vec{\omega}^\nu$	$\vec{\beta} = \begin{bmatrix} \gamma \\ \gamma\boldsymbol{\beta} \end{bmatrix}$
$\vec{\mathcal{S}}_\rho = \dfrac{\partial \vec{x}}{\partial \rho} = U^\nu \vec{\mathrm{e}}_\nu$	$\vec{\mathcal{J}}^\rho = \partial_\nu \rho \vec{\omega}^\nu = -U_\nu \vec{\omega}^\nu$	$\vec{U} = \dfrac{1}{\gamma(1-\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}})} \begin{bmatrix} 1 \\ \hat{\boldsymbol{n}} \end{bmatrix} - \begin{bmatrix} \gamma \\ \gamma\boldsymbol{\beta} \end{bmatrix}$
$\vec{\mathcal{S}}_\theta = \dfrac{\partial \vec{x}}{\partial \theta} = R\,\vartheta^\nu \vec{\mathrm{e}}_\nu$	$\vec{\mathcal{J}}^\theta = \partial_\nu \theta \vec{\omega}^\nu = -\dfrac{1}{R}\vartheta_\nu \vec{\omega}^\nu$	$\vec{\vartheta} = \dfrac{1}{(1-\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}})} \begin{bmatrix} \boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}} \\ \hat{\boldsymbol{\theta}} + \boldsymbol{\beta}\times\hat{\boldsymbol{\phi}} \end{bmatrix}$
$\vec{\mathcal{S}}_\phi = \dfrac{\partial \vec{x}}{\partial \phi} = R\sin\theta\,\varphi^\nu \vec{\mathrm{e}}_\nu$	$\vec{\mathcal{J}}^\phi = \partial_\nu \phi \vec{\omega}^\nu = -\dfrac{1}{R\sin\theta}\varphi_\nu \vec{\omega}^\nu$	$\vec{\varphi} = \dfrac{1}{(1-\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}})} \begin{bmatrix} \boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}} \\ \hat{\boldsymbol{\phi}} - \boldsymbol{\beta}\times\hat{\boldsymbol{\theta}} \end{bmatrix}$

Table 1: *Covariant, contravariant, and unit basis vectors in the spherically-based coordinate system.*

Abstract: While the spherical basis is written specifically for application to classical electron theory, the coordinate system itself may be applied to any theory in physics designed around special relativity. In this investigation, the theory of free Dirac particles will be cast into the spherical basis. Once completed, a spherical basis Klein-Gordon equation will be derived from a suitable operation on the Dirac equation.

2 Dirac Particles in the Spherical Basis

The problem of a Dirac particle can be addressed by projecting the set of Cartesian gamma matrices onto the spherical basis. For this, write the Cartesian gamma matrices and γ^5 as

$$\gamma^0 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \qquad \boldsymbol{\gamma} = \begin{bmatrix} 0 & \boldsymbol{\sigma} \\ -\boldsymbol{\sigma} & 0 \end{bmatrix} \qquad \gamma^5 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \tag{2.1}$$

This is the Dirac Representation of the γ-matrices which will be referenced on several occasions.

2.1 Gamma-Matrices in the Spherical Basis

An interesting possibility—made available by the four-vectors in Table 1—is the ability to project Dirac gamma matrices onto the spherical basis. Using all eight four-vectors from the table, write

$$\vec{\gamma} = (\vec{\gamma} \cdot \vec{\mathcal{J}}^{\tau})\vec{\mathcal{S}}_{\tau} + (\vec{\gamma} \cdot \vec{\mathcal{J}}^{\rho})\vec{\mathcal{S}}_{\rho} + (\vec{\gamma} \cdot \vec{\mathcal{J}}^{\theta})\vec{\mathcal{S}}_{\theta} + (\vec{\gamma} \cdot \vec{\mathcal{J}}^{\phi})\vec{\mathcal{S}}_{\phi} \tag{2.2}$$

Evaluating individual dot products allows this construction to be written in terms of unit vectors in the spherical system

$$\vec{\gamma} = (\gamma^{\alpha}\beta_{\alpha})\vec{\beta} - (\gamma^{\alpha}U_{\alpha})\,\vec{U} - (\gamma^{\alpha}\vartheta_{\alpha})\vec{\vartheta} - (\gamma^{\alpha}\varphi_{\alpha})\vec{\varphi} \tag{2.3}$$

Now suppose components of (2.3) are collected together as

$$\gamma^{\nu} \equiv [\gamma^{\tau}, \gamma^{\rho}, \gamma^{\theta}, \gamma^{\phi}] = [\gamma^{\nu}\beta_{\nu},\ -\gamma^{\nu}U_{\nu},\ -\gamma^{\nu}\vartheta_{\nu},\ -\gamma^{\nu}\varphi_{\nu}] \tag{2.4}$$

Similar to the Cartesian gamma matrices, the sphercially based gamma matrices form an orthogonal set of basis vectors for Minkowski space with norms $[\gamma^{\tau}]^2 = 1$ and $[\gamma^{\rho}]^2 = [\gamma^{\theta}]^2 = [\gamma^{\phi}]^2 = -1$. Evaluating all possible anti-commutation relations determines components of the flat spacetime metric tensor from

$$\{\gamma^{\mu}, \gamma^{\nu}\} = 2\mathrm{I}_4 \cdot \mathrm{g}^{\mu\nu} \tag{2.5}$$

This equation requires the individual elements γ^{ν} to be viewed as generators of the Clifford algebra $C\ell_{1,3}(C)$, except determined in the spherical basis. The validity of this assertion is confirmed by construction of the matrix γ^5 and the related anti-commutation relations:

$$\gamma^5 = i\gamma^{\tau}\gamma^{\rho}\gamma^{\theta}\gamma^{\phi} \qquad \text{and} \qquad \{\gamma^5,\, \gamma^{\nu}\} = 0 \tag{2.6}$$

A concern arising from (2.3) is the notion that each γ-matrix is a function of the particle four velocity in the spherical basis. This is very counter-intuituve and contrary to the (familiar) frame-independent composition of γ-matrices using only the set of numbers $(1, i)$. While this concern is understandable, the mathematics of the spherical basis is completely legitimate, and equations (2.2) through (2.6) can be verified in each of the chiral, Dirac, and Majorana representations of the Dirac algebra. Moreover, it can be expected that contraction theorems, trace theorems, and any other functions involving γ-matrices will still hold in the spherical basis.

Reduction to the Radiation Basis: An important operation is the expansion of equation (2.3) in terms of radiation basis vectors $\{\hat{\boldsymbol{t}}, \hat{\boldsymbol{n}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}\}$. Copius cancellations on the right side reduce the γ-matrices to

$$\vec{\gamma} = \gamma^0 \hat{\boldsymbol{t}} + (\boldsymbol{\gamma} \cdot \hat{\boldsymbol{n}})\hat{\boldsymbol{n}} + (\boldsymbol{\gamma} \cdot \hat{\boldsymbol{\theta}})\hat{\boldsymbol{\theta}} + (\boldsymbol{\gamma} \cdot \hat{\boldsymbol{\phi}})\hat{\boldsymbol{\phi}} \tag{2.7}$$

This defines the set of four radiation basis γ-matrices:

$$\gamma^\nu \equiv [\gamma^0, \boldsymbol{\gamma} \cdot \hat{\boldsymbol{n}}, \boldsymbol{\gamma} \cdot \hat{\boldsymbol{\theta}}, \boldsymbol{\gamma} \cdot \hat{\boldsymbol{\phi}}] \tag{2.8}$$

The emergence of this "hidden" set is intimately related to the design of the spherically based coordinate system—but it doesn't mean the γ-matrices of (2.4) have been overspecified, and it does not diminish their usefulness. Regardless, the set in equation (2.8) have norms consistent with the $(+---)$ convention of special relativity. They also still satisfy equations (2.5) and (2.6), which qualifies (2.8) as an additional set of generators for the Clifford algebra $C\ell_{1,3}(C)$. Table 2 below is a summary of different sets of gamma matrices in each of the three coordinate systems. The essential rule of

	Dirac	**Weyl**	**Majorana**
Cartesian Basis	$\vec{\gamma}_{CD}$	$\vec{\gamma}_{CW}$	$\vec{\gamma}_{CM}$
Spherical Basis	$\vec{\gamma}_{SD}$	$\vec{\gamma}_{SW}$	$\vec{\gamma}_{SM}$
Radiation Basis	$\vec{\gamma}_{RD}$	$\vec{\gamma}_{RW}$	$\vec{\gamma}_{RM}$

Table 2: *Nine sets of gamma-matrices for the Dirac algebra determined with the inclusion of spherically based coordinate systems.*

thumb—for each of nine possible sets—is to apply them to an arbitrary classical tensor with components defined in the same coordinate system as the gamma-matrices.

Unitary Transformations: The transformation law connecting the spherically based gamma matrices to the Cartesian gamma matrices —for any of the three representations—can be written[1]

$$\begin{bmatrix} \gamma^\tau \\ -\gamma^\rho \\ -\gamma^\theta \\ -\gamma^\phi \end{bmatrix} = \begin{bmatrix} \beta_0 & \beta_x & \beta_y & \beta_z \\ U_0 & U_x & U_y & U_z \\ \vartheta_0 & \vartheta_x & \vartheta_y & \vartheta_z \\ \varphi_0 & \varphi_x & \varphi_y & \varphi_z \end{bmatrix} \cdot \begin{bmatrix} \gamma^0 \\ \gamma^x \\ \gamma^y \\ \gamma^z \end{bmatrix} \tag{2.10}$$

The inverse transformation is,

$$\begin{bmatrix} \gamma^0 \\ \gamma^x \\ \gamma^y \\ \gamma^z \end{bmatrix} = \begin{bmatrix} \beta_0 & -U_0 & -\theta_0 & -\phi_0 \\ -\beta_x & U_x & \theta_x & \phi_x \\ -\beta_y & U_y & \theta_y & \phi_y \\ -\beta_z & U_z & \theta_z & \phi_z \end{bmatrix} \cdot \begin{bmatrix} \gamma^\tau \\ -\gamma^\rho \\ -\gamma^\theta \\ -\gamma^\phi \end{bmatrix} \tag{2.11}$$

Similarity transformations among representations in the spherical basis are also possible. As an illustration, consider a transformation of gamma matrices from the Dirac representation to the Weyl representation in the Cartesian coordinate system:

$$\gamma^\nu_{\scriptscriptstyle W} = \mathcal{U}\gamma^\nu_{\scriptscriptstyle D}\mathcal{U}^\dagger \quad \text{where} \quad \mathcal{U} = \frac{1}{\sqrt{2}}\begin{bmatrix} \mathrm{I}_2 & -\mathrm{I}_2 \\ \mathrm{I}_2 & \mathrm{I}_2 \end{bmatrix} \tag{2.12}$$

But this formula—and the unitary matrix itself—is still valid for a similarity transformation among the spherically based gamma-matrices.

$$\gamma^\tau_{\scriptscriptstyle W} = \mathcal{U}\gamma^\tau_{\scriptscriptstyle D}\mathcal{U}^\dagger \qquad \gamma^\rho_{\scriptscriptstyle W} = \mathcal{U}\gamma^\rho_{\scriptscriptstyle D}\mathcal{U}^\dagger \tag{2.13}$$

$$\gamma^\theta_{\scriptscriptstyle W} = \mathcal{U}\gamma^\theta_{\scriptscriptstyle D}\mathcal{U}^\dagger \qquad \gamma^\phi_{\scriptscriptstyle W} = \mathcal{U}\gamma^\phi_{\scriptscriptstyle D}\mathcal{U}^\dagger \tag{2.14}$$

Pauli σ-Matrices: The spherically based coordinate system can also be applied to the Pauli σ-matrices defined by

$$\vec{\sigma} = \sigma^\nu \vec{e}_\nu \qquad \sigma^\nu = (1, \boldsymbol{\sigma}) \tag{2.15a}$$

$$\vec{\bar{\sigma}} = \bar{\sigma}^\nu \vec{e}_\nu \qquad \bar{\sigma}^\nu = (1, -\boldsymbol{\sigma}) \tag{2.15b}$$

Expansions in the spherical basis are:

$$\vec{\sigma} = (\sigma^\alpha\beta_\alpha)\vec{\beta} - (\sigma^\alpha U_\alpha)\vec{U} - (\sigma^\alpha\vartheta_\alpha)\vec{\vartheta} - (\sigma^\alpha\varphi_\alpha)\vec{\varphi} \tag{2.16a}$$

$$\vec{\bar{\sigma}} = (\bar{\sigma}^\alpha\beta_\alpha)\vec{\beta} - (\bar{\sigma}^\alpha U_\alpha)\vec{U} - (\bar{\sigma}^\alpha\vartheta_\alpha)\vec{\vartheta} - (\bar{\sigma}^\alpha\varphi_\alpha)\vec{\varphi} \tag{2.16b}$$

[1] A similar transformation can also be applied to differentials, but it's easier to apply spherical basis vectors to the four-gradient operator of equation (2.23b) to determine

$$\partial_\tau = \beta^\nu\partial_\nu \qquad \partial_\rho = U^\nu\partial_\nu \qquad \partial_\theta = R\cdot\vartheta^\nu\partial_\nu \qquad \partial_\phi = R\sin\theta\cdot\varphi^\nu\partial_\nu \tag{2.9}$$

This identifies two sets of Pauli matrices which can be written

$$\sigma^\nu \equiv [\sigma^\tau, \sigma^\rho, \sigma^\theta, \sigma^\phi] = [\sigma^\nu\beta_\nu, -\sigma^\nu U_\nu, -\sigma^\nu\vartheta_\nu, -\sigma^\nu\varphi_\nu] \tag{2.17a}$$

$$\bar{\sigma}^\nu \equiv [\bar{\sigma}^\tau, \bar{\sigma}^\rho, \bar{\sigma}^\theta, \bar{\sigma}^\phi] = [\bar{\sigma}^\nu\beta_\nu, -\bar{\sigma}^\nu U_\nu, -\bar{\sigma}^\nu\vartheta_\nu, -\bar{\sigma}^\nu\varphi_\nu] \tag{2.17b}$$

An important anti-commutation relation requiring σ-matrices from both sets is

$$\{\sigma^\mu, \bar{\sigma}^\nu\} = 2\mathrm{I}_2 \cdot \mathrm{g}^{\mu\nu} \tag{2.18}$$

Components of spherically based σ-matrices can also be combined to produce complexified electromagnetic velocity fields. For this, include the dilatation η to produce:

$$\eta\bar{\sigma}^\rho \cdot \sigma^\tau = \boldsymbol{\sigma} \cdot (\mathbf{E} + i\mathbf{B}) = -\eta\bar{\sigma}^\tau \cdot \sigma^\rho \tag{2.19a}$$

$$\eta\sigma^\tau \cdot \bar{\sigma}^\rho = \boldsymbol{\sigma} \cdot (\mathbf{E} - i\mathbf{B}) = -\eta\sigma^\rho \cdot \bar{\sigma}^\tau \tag{2.19b}$$

Similar equations derived from angular components are:

$$\eta\bar{\sigma}^\theta \cdot \sigma^\phi = \boldsymbol{\sigma} \cdot (\mathbf{B} - i\mathbf{E}) = -\eta\bar{\sigma}^\phi \cdot \sigma^\theta \tag{2.20a}$$

$$\eta\sigma^\phi \cdot \bar{\sigma}^\theta = \boldsymbol{\sigma} \cdot (\mathbf{B} + i\mathbf{E}) = -\eta\sigma^\theta \cdot \bar{\sigma}^\phi \tag{2.20b}$$

Each the relations in equations (2.19) and (2.20) are complex roots of the electromagnetic Lagrangian. They can be organized in different combinations to determine $\mathcal{L}_{em}$ up to a constant. Examples using all coordinates of the spherical basis are:

$$i\eta^2\sigma^\tau \cdot \bar{\sigma}^\rho \cdot \sigma^\theta \cdot \bar{\sigma}^\phi = \mathbf{E}^2 - \mathbf{B}^2 \qquad i\sigma^\tau \cdot \bar{\sigma}^\rho \cdot \sigma^\theta \cdot \bar{\sigma}^\phi = \mathrm{I}_2 \tag{2.21a}$$

$$-i\eta^2\bar{\sigma}^\tau \cdot \sigma^\rho \cdot \bar{\sigma}^\theta \cdot \sigma^\phi = \mathbf{E}^2 - \mathbf{B}^2 \qquad -i\bar{\sigma}^\tau \cdot \sigma^\rho \cdot \bar{\sigma}^\theta \cdot \sigma^\phi = \mathrm{I}_2 \tag{2.21b}$$

2.2 Dirac Lagrangian and Equation of Motion

The well-known Lagrangian for a Dirac particle of mass m, interacting with an electromagnetic four-potential is

$$\mathcal{L}_D = \bar{\psi}\left[i\hbar c\gamma^\nu\partial_\nu - mc^2\right]\psi - e\bar{\psi}\gamma^\nu\psi A_\nu \tag{2.22}$$

While $\mathcal{L}_D$ is independent of the set of coordinates, the perferred choice is the Cartesian system. To view it in the spherical basis, first write the wave function as $\psi = \psi(c\tau, \rho, \theta, \phi)$, then replace components of the vector potential and the gradient operator by

$$A_\nu \longrightarrow (\beta^\sigma A_\sigma)\beta_\nu - (U^\sigma A_\sigma)\,U_\nu - (\vartheta^\sigma A_\sigma)\vartheta_\nu - (\varphi^\sigma A_\sigma)\varphi_\nu \tag{2.23a}$$

$$\partial_\nu \longrightarrow \beta_\nu\partial_\tau - U_\nu\partial_\rho - R^{-1}\vartheta_\nu\partial_\theta - (R\sin\theta)^{-1}\varphi_\nu\partial_\phi \tag{2.23b}$$

Inserting these into (2.22) will make for a somewhat lengthy equation, but it can be effectively organized into individual parts associated with each spherical coordinate by writing

$$\mathcal{L}_D \equiv \mathcal{L}_\tau + \mathcal{L}_\rho + \mathcal{L}_\theta + \mathcal{L}_\phi \tag{2.24}$$

with individual portions

$$\mathcal{L}_\tau = i\hbar c\bar{\psi}\left[\gamma^\nu\beta_\nu\partial_\tau\right]\psi - e\bar{\psi}\gamma^\nu\beta_\nu\psi\cdot\beta^\sigma A_\sigma - mc^2\bar{\psi}\psi \tag{2.25a}$$

$$\mathcal{L}_\rho = -i\hbar c\bar{\psi}\left[\gamma^\nu U_\nu\partial_\rho\right]\psi + e\bar{\psi}\gamma^\nu U_\nu\psi\cdot U^\sigma A_\sigma \tag{2.25b}$$

$$\mathcal{L}_\theta = -i\hbar c\bar{\psi}\left[R^{-1}\gamma^\nu\vartheta_\nu\partial_\vartheta\right]\psi + e\bar{\psi}\gamma^\nu\vartheta_\nu\psi\cdot\vartheta^\sigma A_\sigma \tag{2.25c}$$

$$\mathcal{L}_\phi = -i\hbar c\bar{\psi}\left[(R\sin\theta)^{-1}\gamma^\nu\varphi_\nu\partial_\varphi\right]\psi + e\bar{\psi}\gamma^\nu\varphi_\nu\psi\cdot\varphi^\sigma A_\sigma \tag{2.25d}$$

Defining the scalar $\lambda \equiv \lambda(c\tau, \rho, \theta, \phi)$, this Lagrangian is still invariant under a local gauge transformation,

$$\psi \longrightarrow \psi\,\mathrm{e}^{-i\alpha\lambda} \qquad A_\nu \longrightarrow A_\nu + \alpha\,\partial_\nu\lambda \qquad \text{with} \qquad \alpha = e/\hbar c \tag{2.26}$$

but a stronger statement can be made as each contribution to the total Lagrangian in (2.24) is—by itself—invariant under a local gauge transformation. This property can be understood by writing the differential

$$\partial_\nu\lambda = \frac{\partial\lambda}{\partial c\tau}\cdot\beta_\nu - \frac{\partial\lambda}{\partial\rho}\cdot U_\nu - \frac{1}{R}\frac{\partial\lambda}{\partial\theta}\cdot\vartheta_\nu - \frac{1}{R\sin\theta}\frac{\partial\lambda}{\partial\phi}\cdot\varphi_\nu \tag{2.27}$$

The covariant derivative still exists in the spherical basis but is more difficult to display in a simple closed form. Instead, individual components may be derived from components of $\mathcal{L}_D$ above:

$$D_{c\tau} = \partial_{c\tau} + \frac{ie}{\hbar c}\beta^\sigma A_\sigma \qquad D_\rho = \partial_\rho + \frac{ie}{\hbar c}U^\sigma A_\sigma \tag{2.28a}$$

$$D_\theta = R^{-1}\partial_\theta + \frac{ie}{\hbar c}\vartheta^\sigma A_\sigma \qquad D_\phi = (R\sin\theta)^{-1}\partial_\phi + \frac{ie}{\hbar c}\varphi^\sigma A_\sigma \tag{2.28b}$$

A more compact form of $\mathcal{L}_D$ follows as

$$\mathcal{L}_D = i\hbar c\bar{\psi}\left[\gamma^\nu\beta_\nu D_{c\tau} - \gamma^\nu U_\nu D_\rho - \gamma^\nu\vartheta_\nu D_\theta - \gamma^\nu\varphi_\nu D_\phi\right]\psi - mc^2\bar{\psi}\psi \tag{2.29}$$

The equation of motion associated with $\mathcal{L}_D$ can be read directly from (2.29) by simply removing the conjugate field $\bar{\psi}$. Instead, it is more practical to remove $\bar{\psi}$ from (2.25) and add individual components. This determines

$$0 = i\hbar c\left[\gamma^\nu\beta_\nu\partial_\tau\right]\psi - e\gamma^\nu\beta_\nu\psi\cdot\beta^\sigma A_\sigma - mc^2\psi$$

$$- i\hbar c\left[\gamma^\nu U_\nu\partial_\rho\right]\psi + e\gamma^\nu U_\nu\psi\cdot U^\sigma A_\sigma$$

$$- i\hbar c\left[R^{-1}\gamma^\nu\vartheta_\nu\partial_\vartheta\right]\psi + e\gamma^\nu\vartheta_\nu\psi\cdot\vartheta^\sigma A_\sigma$$

$$- i\hbar c\left[(R\sin\theta)^{-1}\gamma^\nu\varphi_\nu\partial_\varphi\right]\psi + e\gamma^\nu\varphi_\nu\psi\cdot\varphi^\sigma A_\sigma \tag{2.30}$$

2.3 Solutions to the Dirac Equation in the Spherically Based System

There are many simple possibilities in the spherical basis discussed in each of the six examples below:

Example 1: Free Particle: For the free electron, assume the solution depends only on the time coordinate, yielding the simplified Lagrangian

$$\mathcal{L}_D = \bar\psi\left[i\hbar c\gamma^\nu\beta_\nu\partial_\tau - mc^2\right]\psi \tag{2.31}$$

Equations of motion for both ψ and $\bar\psi$ are

$$\frac{\partial\mathcal{L}_D}{\partial\bar\psi} = 0 \qquad\qquad i\hbar c\gamma^\nu\beta_\nu\partial_\tau\psi - mc^2\psi = 0 \tag{2.32a}$$

$$\partial_\tau\frac{\partial\mathcal{L}_D}{\partial(\partial_\tau\psi)} - \frac{\partial\mathcal{L}_D}{\partial\psi} = 0 \qquad\qquad i\hbar c\partial_\tau\bar\psi\gamma^\nu\beta_\nu + mc^2\bar\psi = 0 \tag{2.32b}$$

The solution for (2.32a) is

$$\psi(\tau) = \psi_o\cdot\mathrm{e}^{\pm imc^2\tau/\hbar} \tag{2.33}$$

Inserting into the Dirac equation determines momentum space equations for electrons and positrons

$$\left[\gamma^\nu p_\nu \pm mc^2\right]\psi_o = 0 \tag{2.34}$$

In other words, ψ_o is not altered by the spherical basis. Now suppose the argument of the exponent in equation (2.33) is transformed via

$$c\tau = \gamma(ct - \boldsymbol{\beta}\cdot\boldsymbol{x}) \tag{2.35}$$

This represents a transformation of the Dirac spinor back to the Cartesian basis.

Example 2: Noether Current for a Free Particle: From the Lagrangian of equation (2.22), one may identify the Noether current $J^\nu = e\bar{\psi}\gamma^\nu\psi$, with the definition $\bar{\psi} = \psi^\dagger\gamma^0$ unchanged by the spherical basis. Calculation of this current proceeds via:

$$J^\nu = e\left[(\bar{\psi}\gamma^\alpha\beta_\alpha\psi)\beta^\nu - (\bar{\psi}\gamma^\alpha U_\alpha\psi)\,U^\nu - (\bar{\psi}\gamma^\alpha\vartheta_\alpha\psi)\vartheta^\nu - (\bar{\psi}\gamma^\alpha\varphi_\alpha\psi)\varphi^\nu\right] \tag{2.36}$$

With wave functions normalized by $\psi^\dagger\psi = 1$, individual products are

$$\bar{\psi}\gamma^\alpha\beta_\alpha\psi = 1 \qquad\qquad \bar{\psi}\gamma^\alpha U_\alpha\psi = \bar{\psi}\gamma^\alpha\vartheta_\alpha\psi = \bar{\psi}\gamma^\alpha\varphi_\alpha\psi = 0 \tag{2.37}$$

The spacelike terms do not contribute to the current which simplifies to $J^\nu = e\beta^\nu$. This result is identical to the Cartesian basis calculation—satisfying $\partial_\nu J^\nu = 0$. Lastly, one can show that J^ν also follows in terms of the null vector R^ν.

$$J^\nu = \frac{e}{\rho}(\bar{\psi}\gamma^\alpha R_\alpha\psi)\beta^\nu \tag{2.38}$$

Example 3: Free Particle with Gauge interactions: Begin with the scalar field depending linearly on all three spatial coordinates

$$\zeta(\rho,\theta,\phi) = -\frac{\hbar c}{e}\left[\kappa_\rho\cdot\rho + \kappa_\theta\cdot\theta + \kappa_\phi\cdot\phi\right] \tag{2.39}$$

Using Table 1, the gradient of this scalar easily follows

$$\partial_\nu\zeta = A_\nu = \frac{\hbar c}{e}\left[\kappa_\rho U_\nu + \frac{1}{R}\cdot\kappa_\theta\vartheta_\nu + \frac{1}{R\sin\theta}\cdot\kappa_\phi\varphi_\nu\right] \tag{2.40}$$

As the vector potential is the gradient of a scalar, no external fields are involved. The Dirac equation for this problem is all of equation (2.30) with solution

$$\psi(c\tau,\rho,\theta,\phi) = \psi_o\cdot \mathrm{e}^{-i(\omega\tau-\kappa_\rho\cdot\rho-\kappa_\theta\cdot\theta-\kappa_\phi\cdot\phi)} \qquad \text{where} \qquad \omega = mc^2/\hbar \tag{2.41}$$

Inserting into the Dirac equation then leads to the momentum space equation

$$\left[\gamma^\nu p_\nu - mc\right]\psi_o = 0 \tag{2.42}$$

Thus, we can arrive at the free particle momentum-space Dirac equation in the spherical basis even with gauge transformations on the coordinates.

Example 4: Weyl Neutrinos: The problem of Weyl neutrinos arises when the mass term in the Dirac equation is set to zero. As massless particles propagate at light speed, neutrinos can't be a function of only $c\tau$. Instead—with the vector potential everywhere zero—Weyl neutrinos will satisfy equation (2.30) in the form

$$i\hbar c\left[\gamma^\nu\beta_\nu\partial_\tau - \gamma^\nu U_\nu\partial_\rho\right]\psi = 0 \tag{2.43}$$

The solution is a spherical wave given by

$$\psi(\tau,\rho) = \psi_o \cdot \mathrm{e}^{\pm ik(c\tau-\rho)} = \psi_o \cdot \mathrm{e}^{\pm ikct_r/\gamma} \tag{2.44}$$

Inserting into (2.43) and employing the relation $R_\nu = \rho(\beta_\nu + U_\nu)$ determines the momentum space eigenvalue equation:

$$\varepsilon\left[\frac{\gamma^\nu R_\nu}{\rho}\right]\psi_o = 0 \qquad \text{with} \qquad \varepsilon = \hbar ck \tag{2.45}$$

But this equation can be made more precise by writing $R_\nu = Rn_\nu$. With $\rho = R^\nu\beta_\nu$ it assumes the form

$$[\gamma^\nu\varepsilon_\nu]\,\psi_o = 0 \qquad \text{where} \qquad \varepsilon^\nu = \frac{(\varepsilon,\boldsymbol{\varepsilon})}{\gamma(1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta})} = (\varepsilon_r,\boldsymbol{\varepsilon}_r) \tag{2.46}$$

The Doppler shifted energy is expected, but the right side of (2.44) requires Weyl neutrinos to be radiated from a source. This seems to be a good idea since all neutrinos detected in high energy experiments are decay products from muons, pions, neutrons and other particles. On the other hand, there is no indication as to what particle emitted the Weyl neutrino.

Example 5: Noether Current for Weyl Neutrinos: Since Weyl neutrinos do not carry charge or mass, J^ν must be interpreted to have units of energy. It is also useful to re-formulate J^ν in terms of components along the null vector. Using $R^\nu = \rho(\beta^\nu + U^\nu)$ components of the current may be determined from

$$\begin{aligned} J^\nu/\varepsilon_r = (\bar\psi\gamma^\alpha R_\alpha\psi)\frac{\beta^\nu}{\rho} + (\bar\psi\gamma^\alpha\beta_\alpha\psi)\frac{R^\nu}{\rho} \\ - (\bar\psi\gamma^\alpha R_\alpha\psi)\frac{R^\nu}{\rho^2} - (\bar\psi\gamma^\alpha\vartheta_\alpha\psi)\vartheta^\nu - (\bar\psi\gamma^\alpha\varphi_\alpha\psi)\varphi^\nu \end{aligned} \tag{2.47}$$

To evaluate each matrix element, use Dirac Spinors in the Dirac representation with normalizations $\psi^\dagger\psi = 1$. A possible eigenstate is

$$\psi = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 0 \\ n_z \\ n_x + in_y \end{bmatrix} \cdot \mathrm{e}^{-ik(c\tau-\rho)} \tag{2.48}$$

One finds that four of the five matrix elements are zero—reducing the current to

$$J^\nu = \varepsilon_r(\bar\psi\gamma^\alpha\beta_\alpha\psi)\frac{R^\nu}{\rho} = \varepsilon^\nu \tag{2.49}$$

Example 6: New Eigenvalue Problem for Dirac Spinors: Electron velocity fields $(\mathbf{E}, \mathbf{B})$ can be used to define the 4×4 field operator

$$\varDelta_v = \begin{bmatrix} i\boldsymbol{\sigma} \cdot \mathbf{B} & \boldsymbol{\sigma} \cdot \mathbf{E} \\ \boldsymbol{\sigma} \cdot \mathbf{E} & i\boldsymbol{\sigma} \cdot \mathbf{B} \end{bmatrix}_v \tag{2.50}$$

This operator arises when multiplying through equation (2.32) by η times the neutrino operator:

$$\begin{aligned} &\eta \frac{\gamma^\nu R_\nu}{\rho} \left[i\hbar c \gamma^\nu \beta_\nu \partial_\tau - mc^2 \right] \psi \\ &\qquad = i\hbar c \left[\varDelta_v + \eta \right] \partial_\tau \psi - mc^2 \eta \frac{\gamma^\nu R_\nu}{\rho} \psi = 0 \end{aligned} \tag{2.51}$$

The second line is still a free particle Dirac equation, but now linked to operators for electron/positron Coulomb fields and Weyl neutrinos. Inserting known solutions and canceling factors of mc^2 leads to a set of new "momentum space" equations—written next to the standard momentum space equations for clarity:

$$e^- : \qquad \left[\varDelta_v + \eta - \eta \frac{\gamma^\nu R_\nu}{\rho} \right] u = 0 \qquad\qquad [\gamma^\nu p_\nu - mc] u = 0 \tag{2.52a}$$

$$e^+ : \qquad \left[\varDelta_v + \eta + \eta \frac{\gamma^\nu R_\nu}{\rho} \right] v = 0 \qquad\qquad [\gamma^\nu p_\nu + mc] v = 0 \tag{2.52b}$$

The solutions to the electromagnetic equations are still (true blue) Dirac spinors for electons and positrons—but instead—eigenstates of the electromagnetic operator. Now suppose the electromagnetic equations are written

$$[\mathcal{A} - \mathcal{B}] u = 0 \qquad [\mathcal{A} + \mathcal{B}] v = 0 \qquad \begin{cases} \mathcal{A} \equiv \varDelta_v + \eta \\ \mathcal{B} \equiv \eta \dfrac{\gamma^\nu R_\nu}{\rho} \end{cases} \tag{2.53}$$

Applying operators a second time produces the following results

$$e^- : \quad [\mathcal{A} + \mathcal{B}][\mathcal{A} - \mathcal{B}] u = 2\eta [\mathcal{A} - \mathcal{B}] u \qquad [\mathcal{A} - \mathcal{B}][\mathcal{A} - \mathcal{B}] u = 2\eta [\mathcal{A} - \mathcal{B}] u \tag{2.54a}$$

$$e^+ : \quad [\mathcal{A} - \mathcal{B}][\mathcal{A} + \mathcal{B}] v = 2\eta [\mathcal{A} + \mathcal{B}] v \qquad [\mathcal{A} + \mathcal{B}][\mathcal{A} + \mathcal{B}] v = 2\eta [\mathcal{A} + \mathcal{B}] v \tag{2.54b}$$

3 Spherical Wave Dirac Spinor:

A Dirac particle interacts with a vacuum gauge vector potential

$$A_\nu = \frac{i\hbar c}{e\rho} U_\nu = \partial_\nu \xi \quad \text{where} \quad \xi \equiv -\frac{i\hbar c}{e} \ln \rho \tag{3.1}$$

With no dependence on angular coordinates (θ, ϕ) the Dirac equation for this problem may be written

$$i\hbar c \left[\gamma^\nu \beta_\nu \partial_\tau - \gamma^\nu U_\nu \partial_\rho\right] \psi - mc^2 \psi - i\hbar c \frac{1}{\rho} \gamma^\nu U_\nu \psi = 0 \tag{3.2}$$

The solution has similarities to a spherical wave

$$\psi(c\tau, \rho) = \frac{\psi_o}{\rho} \cdot \mathrm{e}^{-i(\omega\tau - \kappa\rho)} \qquad \text{where} \qquad c\kappa < \omega \tag{3.3}$$

Inserting above (which cancels the interaction term) leaves the momentum space equation,

$$\left[E\gamma^\nu \beta_\nu + cp\gamma^\nu U_\nu\right] \psi_o - mc^2 \psi_o = 0 \quad \text{where} \quad \begin{cases} E = \hbar\omega \\ p = \hbar k \end{cases} \tag{3.4}$$

The significance of this problem can be demonstrated by re-application of the operator

$$\left[E\gamma^\nu \beta_\nu + cp\gamma^\nu U_\nu\right] \cdot \left[E\gamma^\nu \beta_\nu + cp\gamma^\nu U_\nu\right] \psi_o - mc^2 \cdot \left[E\gamma^\nu \beta_\nu + cp\gamma^\nu U_\nu\right] \psi_o = 0 \tag{3.5}$$

The anti-commutator is $\{\gamma^\nu \beta_\nu, \gamma^\nu U_\nu\} = 0$, so that (3.5) generates the Einstein relation. A more complete solution follows by first expanding $\gamma^\nu U_\nu$ allowing it to be written

$$\left[(E - cp)\gamma^\nu \beta_\nu + cp\frac{\gamma^\nu R_\nu}{\rho}\right] \psi_o = mc^2 \psi_o \tag{3.6}$$

In a 2×2 matrix format one may write

$$\begin{bmatrix} \mathcal{E} & -\boldsymbol{\sigma} \cdot \boldsymbol{\mathcal{P}} c \\ \boldsymbol{\sigma} \cdot \boldsymbol{\mathcal{P}} c & -\mathcal{E} \end{bmatrix} \psi_o = mc^2 \psi_o \quad \text{where} \quad \begin{cases} \mathcal{E} \equiv \gamma(E - cp) + cp\dfrac{R}{\rho} \\ \boldsymbol{\mathcal{P}} c \equiv \gamma(E - cp)\boldsymbol{\beta} + cp\dfrac{R}{\rho}\,\hat{\boldsymbol{n}} \end{cases} \tag{3.7}$$

The eigenvalue mc^2 is already known for this problem, but requires verification to ensure the eigenvalue problem is legitimate. Taking a simple determinant leads to $\lambda^2 = \mathcal{E}^2 - c^2\mathcal{P}^2 = m^2c^4$. Explicit forms of the eigenstates can be demonstrated by writing the matrix equation

$$\begin{bmatrix} \mathcal{E} & -\boldsymbol{\sigma} \cdot \boldsymbol{\mathcal{P}} \\ \boldsymbol{\sigma} \cdot \boldsymbol{\mathcal{P}} & -\mathcal{E} \end{bmatrix} \cdot \begin{bmatrix} \mathcal{E} \pm mc^2 & -\boldsymbol{\sigma} \cdot \boldsymbol{\mathcal{P}} \\ \boldsymbol{\sigma} \cdot \boldsymbol{\mathcal{P}} & -\mathcal{E} \pm mc^2 \end{bmatrix} = \pm mc^2 \begin{bmatrix} \mathcal{E} \pm mc^2 & -\boldsymbol{\sigma} \cdot \boldsymbol{\mathcal{P}} \\ \boldsymbol{\sigma} \cdot \boldsymbol{\mathcal{P}} & -\mathcal{E} \pm mc^2 \end{bmatrix} \tag{3.8}$$

In other words, the columns of the matrix on the right represent legitimate eigenstates of the eigenvalue problem. From example 5, the first of two positive energy eigenstates may be written

$$\psi(c\tau, \rho) = \frac{\mathrm{N}}{\sqrt{4\pi}\rho} \begin{bmatrix} \mathcal{E} + mc^2 \\ 0 \\ c\mathcal{P}_z \\ c\mathcal{P}_x + ic\mathcal{P}_y \end{bmatrix} \cdot \mathrm{e}^{-i(\omega\tau - \kappa\rho)} \tag{3.9}$$

With normalization $\mathrm{N} = 1/\sqrt{2(\mathcal{E}+mc^2)}$ the Noether current $J^\nu = \bar{\psi}\gamma^\nu\psi$ has units of energy per square meter. While J^ν is easily determined using Cartesian gamma matrices, it is instructive perform calculations in the spherical basis. Write

$$J^\nu = (\bar{\psi}\gamma^\nu\beta_\nu\psi)\beta^\nu - (\bar{\psi}\gamma^\nu U_\nu\psi)\,U^\nu = \frac{1}{4\pi\rho^2}\left[(E\beta^\nu + cp\,U^\nu\right] = \frac{1}{4\pi\rho^2}(\mathcal{E}, c\boldsymbol{\mathcal{P}}) \tag{3.10}$$

This formula features two notable limits. When the background velocity $\boldsymbol{\beta}\to\mathbf{0}$, we are operating in the rest frame with the electron moving in a radial direction. The Noether current and its magnitude are

$$J^\nu \longrightarrow \frac{1}{4\pi r^2}(E, cp\hat{\boldsymbol{n}}) \qquad\qquad \sqrt{J^\nu J_\nu} = \frac{mc^2}{4\pi r^2} \tag{3.11}$$

The second limit is when the particle mass is zero. In this scenario the electron becomes a Weyl neutrino satisfying $E = cp$ and chracterized by the current

$$J^\nu = \frac{1}{4\pi\rho^3}R^\nu \tag{3.12}$$

The background velocity $\boldsymbol{\beta}$ is now associated with some unknown particle which emitted the Weyl neutrino. To close, it can be shown that the Noether currents in this problem are divergenceless, except for δ-functions when the radial coordinate tends to zero.

Proper Time Hamiltonian: Multiplication through equation (3.2) by $\eta\cdot\gamma^\nu\beta_\nu$ defines—what looks to be—a Hamiltonian operator equation, except for the proper time derivative:

$$\mathcal{H}\psi \equiv i\hbar\eta\cdot\partial_\tau\psi \qquad \text{where} \qquad \mathcal{H} \equiv -i\hbar c\varDelta_v\left[\partial_\rho + \frac{1}{\rho}\right] + \eta\cdot c\gamma^\nu p_\nu \tag{3.13}$$

The 4×4 matrix $\varDelta_v$ is composed of charged particle velocity fields $(\mathbf{E},\mathbf{B})$ and defined by

$$\varDelta_v \equiv -\eta\cdot\gamma^\nu\beta_\nu\gamma^\mu U_\mu = \begin{bmatrix} i\boldsymbol{\sigma}\cdot\mathbf{B} & \boldsymbol{\sigma}\cdot\mathbf{E} \\ \boldsymbol{\sigma}\cdot\mathbf{E} & i\boldsymbol{\sigma}\cdot\mathbf{B}\end{bmatrix} \tag{3.14}$$

Its square is proportional to the Lagrangian for electromagnetism. Inserting the solution determines the momentum-space eigenvalue problem

$$\mathcal{H}_o\psi \equiv \left[cp\varDelta_v + \eta\gamma^\nu p_\nu\right]\psi_o = \eta E\psi_o \tag{3.15}$$

Application of $\mathcal{H}_o$ again will determine the Einstein relation $p^\nu p_\nu = m^2c^4$.

4 Klein-Gordon Equation

In retarded Minkowski space, associated scale factors can be used to derive the wave operator

$$\Box^2\psi = \frac{\partial^2\psi}{\partial c\tau^2} - \frac{\partial^2\psi}{\partial\rho^2} - \frac{2}{\rho}\frac{\partial\psi}{\partial\rho} - \frac{1}{R^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) - \frac{1}{R^2\sin^2\theta}\cdot\frac{\partial^2\psi}{\partial\phi^2} \tag{4.1}$$

The Klein-Gordon equation along with the $\ell = 0$ solution $\psi(c\tau, \rho)$ are

$$\Box^2\psi + \frac{m^2c^4}{\hbar^2}\psi = 0 \qquad\qquad \psi(c\tau,\rho) = \frac{\psi_o}{\rho}\mathrm{e}^{i(\omega\tau - k\rho)} \tag{4.2}$$

When inserted into (4.1), the solution generates the Einstein energy-momentum relation disguised as a relativistic dispersion relation

$$\omega = \sqrt{c^2k^2 + \frac{m^2c^4}{\hbar^2}} \tag{4.3}$$

The basis independent Noether current for the Klein-Gordon equation reads

$$J^\nu = \frac{i\hbar}{2m}\left[\psi^*\partial^\nu\psi - \psi\,\partial^\nu\psi^*\right] \qquad \text{with} \qquad \partial_\nu J^\nu = 0 \tag{4.4}$$

Inserting the solution once again, and using $E = \hbar\omega$ and $p = \hbar k$, the Noether current and its norm are

$$J^\nu = \frac{|\psi_o|^2}{m\rho^2}\left[E\beta^\nu + cpU^\nu\right] = |\psi_o|^2\left[\frac{\hbar k^\nu}{m\rho^2}\right] \qquad \sqrt{J^\nu J_\nu} = \frac{|\psi_o|^2}{\rho^2} \tag{4.5}$$

Verifying $\partial_\nu J^\nu = 0$ is subtle, requiring application of all terms of the differential operator ∂_ν in the spherical basis.

Derivation from the Dirac Equation: An important question is whether the Klein-Gordon equation has a square root in the spherical basis. The answer is yes, but it's easier to Derive the Klein-Gordon equation by operating on the Dirac equation (2.30). First define the Differential Dirac operator $\mathfrak{D}$

$$\mathfrak{D} = i\hbar c\left[\gamma^\nu\beta_\nu\partial_{c\tau} - \gamma^\nu U_\nu\partial_\rho - \frac{1}{R}\gamma^\nu\vartheta_\nu\partial_\vartheta - \frac{1}{R\sin\theta}\gamma^\nu\varphi_\nu\partial_\varphi\right] \tag{4.6}$$

In the absence of a vector potential, the Dirac equation is $\mathfrak{D}\psi - mc^2\psi = 0$. Now re-apply $\mathfrak{D}$:

$$\mathfrak{D}\cdot\left[\mathfrak{D}\psi - mc^2\psi\right] = \mathfrak{D}^2\psi - m^2c^4\psi = 0 \tag{4.7}$$

Evaluation of $\mathfrak{D}^2$ is

$$\mathfrak{D}^2 = \partial^2_{c\tau} - \partial^2_\rho - \frac{1}{R^2}\partial^2_\theta - \frac{1}{R^2\sin^2\theta}\partial^2_\phi \quad + \quad \text{Six More Terms} \tag{4.8a}$$

Two of the additional terms are not cross terms and require knowledge of derivatives of R with respect to the spacelike variables:

$$\frac{\partial R}{\partial \rho} = \frac{R}{\rho} \qquad \frac{\partial R}{\partial \theta} = \frac{\gamma R^2}{\rho}\boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}} \qquad \frac{\partial R}{\partial \phi} = \frac{\gamma R^2}{\rho}\sin\theta\boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}} \tag{4.8b}$$

These can be used to evaluate

$$\frac{\vartheta^\nu}{R}\partial_\theta\left[\frac{\vartheta_\nu}{R}\right]\partial_\theta = \frac{\gamma\boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}}}{\rho R}\partial_\theta \qquad \frac{\varphi^\nu}{R\sin\theta}\partial_\phi\left[\frac{\varphi_\nu}{R\sin\theta}\right]\partial_\phi = \frac{\gamma\boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}}}{\rho R\sin\theta}\partial_\phi \tag{4.8c}$$

From the four cross terms, two responsible for the third term on the right side of equation (4.1) are

$$\frac{\vartheta^\nu}{R}\cdot[\partial_\theta U_\nu]\partial_\rho = -\frac{1}{\rho}\partial_\rho \qquad \frac{\varphi^\nu}{R\sin\theta}\cdot[\partial_\phi U_\nu]\partial_\rho = -\frac{1}{\rho}\partial_\rho \tag{4.8d}$$

The two remaining cross terms are more difficult to evaluate as they require differentiating the unit vectors ϑ_ν and φ_ν:

$$\frac{\vartheta^\nu}{R}\cdot\partial_\theta\left[\frac{\varphi_\nu}{R\sin\theta}\right]\partial_\phi = -\frac{\gamma\boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}}}{\rho R\sin\theta}\partial_\phi \tag{4.8e}$$

$$\frac{\varphi^\nu}{R\sin\theta}\cdot\partial_\phi\left[\frac{\vartheta_\nu}{R}\right]\partial_\theta = -\left[\frac{\cot\theta}{R^2} + \frac{\gamma\boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}}}{\rho R}\right]\partial_\theta \tag{4.8f}$$

Cancellations among the remaining six terms are now evident, and the final result is the spherical basis Klein-Gordon equation.

A Eigenvalue Problem for Gamma-Matrices

Each γ-matrix in the spherical system $(\gamma^\tau, \gamma^\rho, \gamma^\theta, \gamma^\phi)$ gives rise to its own eigenvalue problem. This problem has already been solved for γ^τ which includes the particle mass m to produce momentum space Dirac equations

$$(\gamma^\nu p_\nu \pm mc)\psi = 0 \tag{A.1}$$

The spacelike γ-matrices also present their own eigenvalue problem with purely imaginary eigenvalues:

$$\gamma^\nu U_\nu \cdot \psi_u = \pm i\, \psi_u \tag{A.2a}$$
$$\gamma^\nu \vartheta_\nu \cdot \psi_\theta = \pm i\, \psi_\theta \tag{A.2b}$$
$$\gamma^\nu \varphi_\nu \cdot \psi_\phi = \pm i\, \psi_\phi \tag{A.2c}$$

If the same operator is applied to both sides of each equation above, one immediately concludes that

$$[\gamma^\nu U_\nu]^2 = [\gamma^\nu \vartheta_\nu]^2 = [\gamma^\nu \varphi_\nu]^2 = -\mathrm{I} \tag{A.3}$$

The eigenvalue problem for the operator equations in (A.2) can all be solved simultaneously. Using the variable $\boldsymbol{w}^\nu$ to represent either four-vector the essential problem in the Dirac representation is

$$\begin{bmatrix} w_o & -\boldsymbol{\sigma}\cdot\boldsymbol{w} \\ \boldsymbol{\sigma}\cdot\boldsymbol{w} & -w_o \end{bmatrix} \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix} = \pm i \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix} \tag{A.4}$$

Eigenstates are similar to Dirac particle eigenstates

$$|+i\rangle_1 = \begin{bmatrix} w_o + i \\ 0 \\ w_z \\ w_x + i w_y \end{bmatrix} \qquad |+i\rangle_2 = \begin{bmatrix} 0 \\ w_o + i \\ w_x - i w_y \\ -w_z \end{bmatrix} \tag{A.5}$$

$$|-i\rangle_1 = \begin{bmatrix} w_z \\ w_x + i w_y \\ w_o + i \\ 0 \end{bmatrix} \qquad |-i\rangle_2 = \begin{bmatrix} w_x - i w_y \\ -w_z \\ 0 \\ w_o + i \end{bmatrix} \tag{A.6}$$

All eigenstates have zero norm which is represented by the relativistic invariant given by $\langle\, \alpha\, |\gamma^0|\, \alpha\, \rangle = 0$. In addition, eigenstates associated with the same eigenvalue are orthogonal ${}_1\langle\, \pm i\, |\gamma^0|\, \pm i\, \rangle_2 = 0$.

B Tensor Elements in the Spherically Based Coordinate System

Dirac gamma-matrices in the spherical coordinate system generate a tensor basis

Scalar: $\{\,\mathrm{I}\,\}$

Vectors: $\{\,\gamma^\tau,\,\gamma^\rho,\,\gamma^\theta,\,\gamma^\phi\}$

Bi-vectors: $\{\,\gamma^\tau\gamma^\rho,\,\gamma^\tau\gamma^\theta,\,\gamma^\tau\gamma^\phi,\,\gamma^\rho\gamma^\theta,\,\gamma^\rho\gamma^\phi,\,\gamma^\theta\gamma^\phi\,\}$

Tri-vectors: $\{\,\gamma^\tau\gamma^\rho\gamma^\theta,\,\gamma^\tau\gamma^\rho\gamma^\phi,\,\gamma^\tau\gamma^\theta\gamma^\phi,\,\gamma^\rho\gamma^\theta\gamma^\phi\,\}$

Pseudoscalar: $\{\,\gamma^\tau\gamma^\rho\gamma^\theta\gamma^\phi\,\}$

For comparison, Table 3 indicates relations among bi-vectors and tri-vectors in the Cartesian and Spherical basis, which only requires a re-labelling the indices.

Bi-Vectors	
Cartesian	Spherical
$i\gamma^1\gamma^2 = \gamma^5\gamma^0\gamma^3$	$i\gamma^\rho\gamma^\theta = \gamma^5\gamma^\tau\gamma^\phi$
$i\gamma^3\gamma^1 = \gamma^5\gamma^0\gamma^2$	$i\gamma^\phi\gamma^\rho = \gamma^5\gamma^\tau\gamma^\theta$
$i\gamma^2\gamma^3 = \gamma^5\gamma^0\gamma^1$	$i\gamma^\theta\gamma^\phi = \gamma^5\gamma^\tau\gamma^\rho$

Tri-Vectors	
Cartesian	Spherical
$\gamma^0\gamma^1\gamma^2 = i\gamma^5\gamma^3$	$\gamma^\tau\gamma^\rho\gamma^\theta = i\gamma^5\gamma^\phi$
$\gamma^0\gamma^3\gamma^1 = i\gamma^5\gamma^2$	$\gamma^\tau\gamma^\phi\gamma^\rho = i\gamma^5\gamma^\theta$
$\gamma^0\gamma^2\gamma^3 = i\gamma^5\gamma^1$	$\gamma^\tau\gamma^\theta\gamma^\phi = i\gamma^5\gamma^\rho$
$\gamma^1\gamma^2\gamma^3 = i\gamma^5\gamma^0$	$\gamma^\rho\gamma^\theta\gamma^\phi = i\gamma^5\gamma^\tau$

Table 3: *Comparison of bi-vectors and tri-vectors in Cartesian and spherical bases.*

Of interest is the specific evaluation of matrices associated with each bi-vector in the spherical system. Inclusion of the dilatation η is almost unavoidable producing electron Coulomb fields in a natural way. Most striking is the relation

$$i\eta\,\gamma^\theta\gamma^\phi = \begin{bmatrix} \boldsymbol{\sigma}\cdot\mathbf{E} & i\boldsymbol{\sigma}\cdot\mathbf{B} \\ i\boldsymbol{\sigma}\cdot\mathbf{B} & \boldsymbol{\sigma}\cdot\mathbf{E} \end{bmatrix} = \eta\,\gamma^5\gamma^\tau\gamma^\rho \tag{B.1}$$

In addition, one may calculate

$$i\eta\,\gamma^\phi\gamma^\rho = \gamma\boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}}\begin{bmatrix} \boldsymbol{\sigma}\cdot\mathbf{E} & i\boldsymbol{\sigma}\cdot\mathbf{B} \\ i\boldsymbol{\sigma}\cdot\mathbf{B} & \boldsymbol{\sigma}\cdot\mathbf{E} \end{bmatrix} + \gamma\eta\begin{bmatrix} \boldsymbol{\sigma}\cdot\hat{\boldsymbol{\theta}} & i\boldsymbol{\sigma}\cdot\boldsymbol{\beta}\times\hat{\boldsymbol{\theta}} \\ i\boldsymbol{\sigma}\cdot\boldsymbol{\beta}\times\hat{\boldsymbol{\theta}} & \boldsymbol{\sigma}\cdot\hat{\boldsymbol{\theta}} \end{bmatrix} = \eta\,\gamma^5\gamma^\tau\gamma^\theta \tag{B.2a}$$

$$i\eta\,\gamma^\theta\gamma^\rho = \gamma\boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}}\begin{bmatrix} \boldsymbol{\sigma}\cdot\mathbf{E} & i\boldsymbol{\sigma}\cdot\mathbf{B} \\ i\boldsymbol{\sigma}\cdot\mathbf{B} & \boldsymbol{\sigma}\cdot\mathbf{E} \end{bmatrix} + \gamma\eta\begin{bmatrix} \boldsymbol{\sigma}\cdot\hat{\boldsymbol{\phi}} & i\boldsymbol{\sigma}\cdot\boldsymbol{\beta}\times\hat{\boldsymbol{\phi}} \\ i\boldsymbol{\sigma}\cdot\boldsymbol{\beta}\times\hat{\boldsymbol{\phi}} & \boldsymbol{\sigma}\cdot\hat{\boldsymbol{\phi}} \end{bmatrix} = \eta\,\gamma^5\gamma^\tau\gamma^\phi \tag{B.2b}$$

Each matrix in equations (B.1) and (B.2) can also be squared to determine the Lagrangian for the electromagnetic field to within a factor of $\pm 1/8\pi$.

C Solutions to the Dirac Equation

the free particle Dirac equation in the spherical basis

$$i\hbar c\left[\gamma^\nu\beta_\nu\frac{\partial}{\partial c\tau}-\gamma^\nu U_\nu\frac{\partial}{\partial\rho}-\gamma^\nu\vartheta_\nu\frac{1}{R}\frac{\partial}{\partial\theta}-\gamma^\nu\varphi_\nu\frac{1}{R\sin\theta}\frac{\partial}{\partial\phi}\right]\psi-mc^2\psi=0 \tag{C.1}$$

Solutions to this equation can be determined beginning with the transformation equation of equation (1.3) with components of $\vec{x}$ given by:

$$x^\nu = c\tau\beta^\nu+\rho U^\nu \tag{C.2}$$

Defining the wave vector $\vec{k}\equiv k^\nu\vec{e_\nu}$, Dirac wave functions may be written:

$$\psi=\psi_o\mathrm{e}^{\pm ik^\nu x_\nu}=\psi_o\,\mathrm{e}^{\pm ic\tau k^\nu\beta_\nu}\cdot\mathrm{e}^{\pm i\rho k^\nu U_\nu} \tag{C.3}$$

The argument of the first exponential is only a function of τ, while the second exponential is a function of all three space variables (ρ,θ,ϕ). Inserting the trial solution into (C.1) determines the algebraic equation

$$\mp\hbar\left[\gamma^\nu\beta_\nu\cdot k^\nu\beta_\nu-\gamma^\nu U_\nu\cdot k^\nu U_\nu-\gamma^\nu\theta_\nu\cdot k^\nu\theta_\nu-\gamma^\nu\phi_\nu\cdot k^\nu\phi_\nu\right]\psi_o=mc\,\psi_o \tag{C.4}$$

But the term in brackets is the dot product $\vec{\gamma}\cdot\vec{k}$ in the spherical system, and this leads to the momentum space Dirac equations

$$\mp\hbar[\gamma^\nu k_\nu]\psi_o=mc\,\psi_o \tag{C.5}$$